
Group 4

SmartSpend AI
Iteration Test Plan: Login & Add Manual Transaction

Version 2.0

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SS-ITP-01	

Revision History

Date	Version	Description	Author
19/07/2026	1.0	Draft	Phan Văn Bá Đạt
13/08/2026	2.0	Change CRUD category to Add Manual Transactions	Phan Văn Bá Đạt

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Iteration Test Plan: Login & Category Management

1. Introduction

This Iteration Test Plan provides a targeted framework to govern the testing activities for the core foundation modules of the **SmartSpend AI** application: the Authentication (Login) engine and the Manage Custom Categories CRUD engine.

- **What are the objectives of this document?**
To establish a distinct verification protocol for testing account access flows and localized relational database modifications within the current project sprint.
- **What is this document about?**
It details the validation of authentication behaviors (standard inputs, third-party Google handshakes, credential validation faults) and the operational evaluation of custom user category manipulation workflows (Create, Read, Update, Delete transactions).
- **Who should use it?**
This document is actively utilized by team quality assurance specialists executing targeted user acceptance simulations and system developers checking state-mutation logic.
- **What are the key features of this document?**
Key features include direct technical expansions of alternative use-case paths into testing targets, specific environment alignments matching our current Supabase/Android architecture, and localized task definitions.

1.1 Purpose

The purpose of this document is to ensure that users can reliably gain access to the system via local or federated identity providers and manipulate personalized tracking scopes without risking local database integrity or structural transaction mapping drops.

2. Target Test Items

2.1 Functional Testing

Login Engine Sub-System:

- **Standard Login:** Verifies entering Email and Password, clicking the "Login" button, and navigating successfully to the Homepage with the logged-in user session.
- **Google Authentication:** Verifies clicking "Login with Google", selecting a Google account, authentication, and navigating to the Homepage.
- **Login Error Handling:** Verifies displaying error dialogs for incorrect or invalid Email/Password inputs, while retaining the entered input fields for correction.

Add Manual Transaction System:

- **View Transaction Records (Read):** Verifies that opening the Add Manual Transaction screen properly displays the blank form with all required fields (Transaction Name, Type, Amount, Date & Time, Category, Note) and that the Category dropdown is populated with both default and the user's custom categories.
- **Create Transaction (Create):** Verifies entering Transaction Name, selecting Type (Income/Expense), entering a valid Amount, choosing Date/Time, selecting a Category from the dropdown, optionally adding a Note, and saving the new transaction to the database. The system should validate all inputs and recalculate the wallet balance.
- **Form Validation Feedback:** Verifies that validation errors are clearly displayed for invalid or missing data (e.g., blank or non-numeric Amount, negative/zero values, missing required fields) so the user can correct them before saving.
- **Save & Sync Impact:** Verifies that after saving, the new transaction immediately updates dependent features (Wallet Balance, Transaction History, Set Monthly Budget allowance, View Financial Reports).

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breakdown) without requiring a manual page reload.

- **Error & Constraint Handling:** Verifies system error pop-ups and retry options when encountering network/database connection timeouts while saving the transaction. The system should prevent data loss or duplication.

2.2 Usability Testing

- Form validation feedback visibility; tap-target spacing for input fields and action buttons (e.g., Save vs. Cancel); responsive layout behaviors of the transaction entry form across varying displays and orientations.

2.3 Performance & Reliability Testing

- **Database & UI Synchronicity (Add Manual Transaction):** Verifies system stability when Supabase connections drop during authentication or while saving a manual transaction. Ensures the app displays graceful error pop-ups, allows retries, and prevents data loss or corruption.

2.4 Reliability Testing

- **Network Failure Resilience (Login & Add Manual Transaction):** Verifies system stability when Supabase connections drop during authentication or while saving a manual transaction. Ensures the app displays graceful error pop-ups, allows retries, and prevents data loss or corruption.

3. Environmental Needs

3.1 Hardware Requirements

- **Host Development Terminal:** Standard x86-64 workstations configured with a multi-core CPU architecture and a minimum of 8GB–16GB RAM to support parallel background emulation pipelines.
- **Physical Client Device Node (Android):** Minimum of one physical Android handheld terminal (operating on Android 11 or later) to execute integration tests under typical system constraint levels.
- **Physical Client Device Node (iOS):** Minimum of one physical Apple iOS device (operating on iOS 15 or later) to validate authentication token handshakes and user layout rendering under native system environments.

3.2 Software in the Test Environment

The following base software elements are required in the test environment for this *Test Plan*.

Software Name	Purpose	Version	Type and Other Notes
Android OS	Mobile operating system	Android 11 or later	Primary testing platform on physical devices to validate camera workflows and native behavior
IOS	Mobile operating system	IOS 15 or later	Secondary testing platform for cross-platform visual layout and compatibility validation
Android Studio	Emulator and development environment	Latest Version	Used for local execution, log inspection, and emulator testing
IOS Simulator	Client layout and structural rendering verification	Included with Xcode	Integrated native emulation platform running within Apple Xcode architectures

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Software Name	Purpose	Version	Type and Other Notes
Google Chrome	Backend/Admin testing	Latest Version	Standard web browser used to access management dashboards and backend configurations
Supabase	Authentication and database services	Latest Version	Cloud backend service hosting user session data and relational financial tables.

3.3 Productivity and Support Tools

The following tools will be employed to support the test process for this *Test Plan*.

Tool Category or Type	Tool Brand Name	Vendor or In-house	Version
Defect & Task Tracking	Jira Software	Vendor	Current Web Interface
Version Control Infrastructure	GitHub Repositories	Vendor	Current Distribution
API Testing & Automation	Custom Test Scripts (VS Code / Python)	In-house	Latest Version

4. Responsibilities, Staffing, and Training Needs

4.1 People and Roles

This table shows the staffing assumptions for the test effort.

Human Resources		
Role	Minimum Resources Recommended (number of full-time roles allocated)	Specific Responsibilities or Comments
Test Manager	1 (Phan Văn Bá Đạt)	Provides management oversight. Responsibilities include: - planning and logistics - agree mission - identify motivators - acquire appropriate resources - present management reporting - advocate the interests of test - evaluate effectiveness of test effort

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Human Resources		
Role	Minimum Resources Recommended (number of full-time roles allocated)	Specific Responsibilities or Comments
Tester	1 (Phan Văn Bá Đạt)	Implements and executes the tests. Responsibilities include: • implement tests and test suites • execute test suites • log results • analyze and recover from test failures • document incidents
Test System Administrator	1 (Phan Văn Bá Đạt)	Ensures test environment and assets are managed and maintained. Responsibilities include: • administer test management system • install and support access to, and recovery of, test environment configurations and test labs